Jakarta Phase B optical feasibility

Scientific supplement • Nabhonil Sarkar • 7 November 2026

Two fixed Sentinel-2 observation designs failed their temporal-support requirements in the first tested season and year. Both stopped before transition classification. The result bounds what these designs could establish; it does not establish universal optical infeasibility or absence of surface change.

Coastal question and fixed designs

Phase B asked whether repeated optical observations could identify persistent land–water transitions in coastal windows around Jakarta during 2020–2025. Adequate observation support had to pass before any land–water classification or interpretation. Four coastal windows (C1–C4), an inland land control (L1) and a water control (W1) were fixed in advance. Each core was nominally 1 km square, containing 2,500 cells on the 20 m projected grid; controls were baseline references, not independently verified unchanged truth. All six windows lie within the report's northwest study sector and were selected geometrically using WorldCover 2020, without using embedding-difference values. [1, 2]

The original design combined January–April and June–September observations in each of the six years: 96 selected scenes and 576 assets. The separate follow-up targeted June–September observed states in the same six years: 48 scenes and 288 assets. It addressed seasonal observations, not year-round permanence. Both retained the same six windows and eight preselected dates per seasonal window, one scene per calendar half-month, chosen using the lowest whole-scene cloud metadata among eligible scenes. Local clear-pixel coverage was not a selection criterion. [1, 2]

The follow-up was prospectively specified on 6 November after the original support failure. Its June–September pixels and transition outcomes had not been inspected when designed. It retained the original June–September scene selections and fixed the acquisition order as 2022 → 2021 → 2024 → 2023 → 2020 → 2025. No later-year pixels could be scheduled unless the current year passed. [2]

Shared observation requirements

A supported cell required at least three clear observations on distinct UTC dates, in at least three distinct months, a first-to-last span of at least 60 days, and at least three dates mutually separated by at least 10 days (a greedy chronological subset). Every window and year-season had to reach 80% of all 2,500 core cells—at least 2,000 cells—under both the 60 m and 120 m quality buffers. Missing observations never counted as clear votes; windows, seasons and buffers could not be pooled. [1, 2]

Clear support used finite calibrated B03/B11 and Sen2Cor scene-classification (SCL) codes 4, 5 or 6, outside buffers around codes 2, 3, 8, 9 and 10. Buffer distances were measured between 20 m cell centres, with the halo processed before core cropping. DN zero was masked before XML-based calibration; aligned B03 averaging required all four native contributors to be valid. SCL exclusions and buffer dilation are quality rules, not proof that every excluded observation was physically cloud-covered. [1, 2]

Technical stop and feasibility stops

The initial attempt stopped before pixels on 5 November at the combined native-header check; the absent TIFF nodata tag was identified as the cause during diagnosis on 6 November. The 6 November diagnosis and tested correction accepted absent or zero TIFF nodata only with verified, unambiguous XML zero semantics. Geometry, calibration, masking and scientific thresholds were unchanged. All 576 headers passed before pixels in corrected attempt 02; it acquired January–April 2020 and stopped at temporal support. All 288 headers passed in the separate seasonal run; it acquired June–September 2022 first and stopped at the same gate. Neither completed its six-year sequence or classified transitions. Seasonal Stage 1 generated no water indices or land–water labels. [3–6]

Measured coverage in all six windows

Entries show cells meeting all temporal rules, their percentage of the fixed 2,500-cell core, and the 80% gate result. Both buffers must pass in every window. These are observation-support measures, not classified land or water area. [4, 5]

Original attempt 02 January to April 2020

Window	60 m support	Gate	120 m support	Gate
C1 coastal	0 (0.00%)	FAIL	0 (0.00%)	FAIL
C2 coastal	0 (0.00%)	FAIL	0 (0.00%)	FAIL
C3 coastal	0 (0.00%)	FAIL	0 (0.00%)	FAIL
C4 coastal	0 (0.00%)	FAIL	0 (0.00%)	FAIL
L1 inland control	2 (0.08%)	FAIL	0 (0.00%)	FAIL
W1 water control	0 (0.00%)	FAIL	0 (0.00%)	FAIL

All 12 measured entries failed. Eight dates completed; the remaining 88 selected dates were unacquired. June–September 2020 and both seasons of 2021–2025 remain unacquired: 132 coverage entries have no measured support value, rather than zero support.

Separate seasonal attempt June to September 2022

Window	60 m support	Gate	120 m support	Gate
C1 coastal	0 (0.00%)	FAIL	0 (0.00%)	FAIL
C2 coastal	2,328 (93.12%)	**PASS**	2,256 (90.24%)	**PASS**
C3 coastal	1,468 (58.72%)	FAIL	1,446 (57.84%)	FAIL
C4 coastal	2,394 (95.76%)	**PASS**	2,369 (94.76%)	**PASS**
L1 inland control	1,277 (51.08%)	FAIL	832 (33.28%)	FAIL
W1 water control	547 (21.88%)	FAIL	547 (21.88%)	FAIL

C2 and C4 passed under both buffers; eight of 12 measured entries failed. Eight dates completed, none were partial, and 40 dates remained unacquired. June–September 2020, 2021, 2023, 2024 and 2025 remain unacquired: 60 coverage entries have no measured support value. Passing individual windows does not establish six-year feasibility.

What constrained temporal support

Both designs used eight preselected dates per seasonal window. After quality masking, retained observations failed the required temporal coverage in multiple windows. The 60-day span was a major constraint, alongside clear-date and month requirements. Using all available scenes or longer seasonal windows was not tested.

The saved counts show why the span alone is not a complete explanation. In January–April 2020, only two L1 cells at 60 m reached a 60-day span; every other window–buffer entry had none. At C1, 2,328 / 2,142 cells had three clear dates, yet none had three months or a 60-day span (60 m / 120 m). In June–September 2022, W1 had 2,422 / 2,268 cells with three clear dates and three months, but only 547 / 547 with a 60-day span. L1 had only 1,277 / 832 cells with three months; C3 had only 1,713 / 1,558 with three clear dates. These marginal counts overlap and do not partition causes of failure. [4, 5]

Resources and evidential limits

Execution	Response-body bytes received	Attempt reserved bytes	Attempt seconds
Original initial	19,652	3,377,970	5.95
Original attempt 02	134,763,696	541,692,466	673.75
Seasonal Stage 1	152,942,030	371,127,124	221.64

Response-body bytes record received payloads; reservations are conservative accounting charges and are not measured traffic. Attempt 02 ended at 814,247,424 cumulative reserved bytes and 733.75 charged seconds, including its prior debit. Seasonal Stage 1 carried that byte debit plus the 67,108,864-byte documentation hold exactly once, ending at 1,252,483,412 bytes and 955.64 seconds (734 seconds carried in conservatively). The 3 GiB accounting ceiling, 7,200-second acquisition ceiling and 7 GiB free-disk gate passed. Historical allowances remain accounting assumptions, not verified lifetime traffic or proven traffic upper bounds. [3–5]

These fixed designs failed their temporal-support requirements at the tested windows and seasons. This does not establish universal optical infeasibility, absence of surface change, or a land-loss, reclamation or subsidence result. No replacement scenes, weaker gates or further optical rescue attempt followed. The optical branch is closed under the current project scope.

The source route used Copernicus Sentinel-2 L2A data via Microsoft Planetary Computer. SCL is an imperfect quality proxy; encoding corrections do not make the mixed processing baselines a uniformly reprocessed archive. Date provenance retains spacecraft, processing baseline and UTC product-start time; those timestamps are not verified per-pixel sensing times. The existing outcome audits document coverage-only stops. No new scientific analysis was performed for this supplement. [2, 4, 5]

Provenance and saved evidence

Paths below are relative to the repository. A denotes data/output/coastal_transition_pilot_v1/ and B denotes data/output/coastal_jun_sep_v1/. Protocol hashes are full SHA-256 identifiers. Run records and audits retain the remaining exact file hashes and acquisition provenance.

[1] A/protocol.json — original fixed two-season design.
SHA-256 4faf2430c6dbce50fb7e69cdcad9db819953bac6bb90edb2de4376cd38698754

[2] B/protocol.json — separate prospective seasonal design, prepared 6 November 2026.
SHA-256 948626ae2e8267177ab8ced5a18ac65f5cef23a019e2cc46fd0c461133a7c917
B/acquisitions.json and stage1_run/coverage.csv retain dated spacecraft/baseline provenance.

[3] A/run/results.json and A/execution_stop_audit.json — initial execution and stop audit, 5 November 2026. Executing commit 5dbdbb4cbae60aeb1513c49ef38c0c797e9f73d5.

[4] A/run_attempt_02/results.json, coverage.csv and coverage.json; A/restart_attempt_02/outcome_audit.json — execution and outcome audit, 6 November 2026. Executing commit 0ac4b05beb456bd2b1212a13b311ef7d02d4bc2f.

[5] B/stage1_run/results.json, coverage.csv and coverage.json; B/stage1_outcome_audit.json — executed 6 November 2026; audited 7 November 2026. The local execution date is recorded from file times in Asia/Singapore. Executing commit b40b916dce012d2b3920afb801f307ad4f967b14.

[6] A/metadata_diagnosis_20261106/diagnosis.json and A/nodata_correction_20261106/verification.json — diagnosis and correction verification, 6 November 2026. The saved correction verification records 305 passing tests; these are historical results, not a new test run.

This results supplement accompanies the accepted Jakarta Surface Differences 2020–2025 report. It does not replace that report or the project's sole methodology.